

Ibrahim Ibrahim

Robotics Software Engineer | Erasmus Mundus MSc Robotics Student

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Personal Statement

Robotics MSc candidate in the Erasmus Mundus programme, expected to graduate in June 2026. Hands-on experience building ROS 2 autonomy pipelines for a Unitree Go2 EDU quadruped, including SLAM-based mapping, localization, motion planning, teleoperation and live robot monitoring website. Seeking Robotics Software / Autonomous Systems roles in Germany.

Education

MSc Erasmus Mundus in Intelligent Field Robotics | University of Girona / University of Zagreb

Girona, Spain / Zagreb, Croatia / Berlin, Germany | September 2024– June 2026

- Erasmus Mundus Joint Master Scholarship recipient; a prestigious, highly intensive and deeply specialized scholarship.
- 1st & 2nd semesters in Spain: field robotics foundations, manipulation, localization, perception, planning and ROS lab projects.
- 3rd semester in Croatia: multi-robot systems, aerial robotics, human-robot interaction and AI for perception
- 4th semester in Germany: internship/ thesis on Autonomous Navigation and control for a Unitree Go2 Quadruped Robot.

BSc Computer Science | University of Pécs

Pécs, Hungary | September 2021– June 2024

- Stipendium Hungaricum Scholarship recipient due to my exceptional academic performance in secondary school.
- Thesis topic: Machine Learning Force Fields for Transition Metal Complexes.
- Notable Modules: Data Analysis, Algorithms and Data Structures, Probability and Statistics, Control Technology.

Work Experience

Robotics Engineer Intern / MSc Thesis | W.I.S. Sicherheit

Berlin, Germany | February 2026 - Present

- Developing a full-stack teleoperation and autonomous navigation system for the Unitree Go2 EDU quadruped robot.
- Implementing ROS 2 C++ pipelines for SLAM-based mapping, localization, motion planning and robot control.
- Building a real-time robot monitoring and control platform using React, Laravel/PHP, WebRTC, WebSockets and NATS.
- Working between robotics back-end, web dashboard and real robot testing to connect autonomy with operator workflows.

Machine Learning Research Intern | CoE Marble

Zagreb, Croatia | July 2025 - October 2025

- Developed and evaluated visual anomaly detection models (Autoencoder U-Net, YOLO, EfficientAD, PatchCore) on offshore and wind-turbine inspection datasets.
- Implemented synthetic anomaly generation pipelines and created a custom underwater inspection dataset.
- Integrated explainability techniques (Grad-CAM, feature attribution maps) to interpret and visualize model decisions.
- Conducted benchmarking and literature review on state-of-the-art visual anomaly detection and explainable AI methods.

Automation & Data Engineer intern | DATelite

Pécs, Hungary | October 2023 – April 2024

- Developed Python ETL pipelines for large-scale earth-observation and geospatial data processing.
- Applied piecewise regression, Kruskal algorithm, discrete optimization, statistical modelling and data visualization to derive meaningful insights from data.

Stipendium Hungaricum Mentor | SHMN

Pécs, Hungary | July 2022 – July 2023

- Welcomed international students to the university and provided up-to-date information about their studies and administrative tasks.
- Demonstrated good organization skills and the ability to plan ahead for 10-15 students.

Portfolio

Visit my online portfolio to explore selected projects in robotics, computer vision, and autonomous systems [🔗 190ibrahim.github.io/My-Portfolio](https://190ibrahim.github.io/My-Portfolio) Includes work on motion planning, SLAM, deep learning for perception, and embedded robotic systems.

Skills

- Robotics: ROS, ROS 2 (ROS2), SLAM, localization, autonomous navigation, motion planning, sensor fusion, Gazebo, RViz
- Programming: C++, Python, Linux/Ubuntu, Git, GitHub, Docker, CMake, SQL
- Perception / AI: OpenCV, PyTorch, YOLO, Grad-CAM, XAI
- Software Development: React, Laravel/PHP, WebRTC, WebSockets, NATS, REST APIs

Languages

- Arabic: Native
- English: Fluent
- Hungarian: Beginner
- German: Beginner